

Q1.GUI using AWT

Code:

```
import java.awt.*;
import java.awt.event.*;

public class RegistrationForm extends Frame implements ActionListener {
    Label titleLabel, subtitleLabel;
    Label personalLabel, academicLabel;
    Label nameLabel, emailLabel, phoneLabel;
    Label passwordLabel, genderLabel, departmentLabel;
    Label yearLabel, skillsLabel, termsLabel;
    Label resultTitleLabel, resultLabel;
    TextField nameField;
    TextField emailField;
    TextField phoneField;
    TextField passwordField;
    Checkbox maleCheckbox;
    Checkbox femaleCheckbox;
    Checkbox javaCheckbox;
    Checkbox pythonCheckbox;
    Checkbox cppCheckbox;
    Checkbox termsCheckbox;
    Choice departmentChoice;
    Choice yearChoice;
    Button registerButton;
    Button clearButton;
    Button exitButton;

    public RegistrationForm() {
        setTitle("Engineering Student Registration System");
        Dimension screen = Toolkit.getDefaultToolkit().getScreenSize();
        int screenWidth = screen.width;
        int screenHeight = screen.height;
        setSize(screenWidth, screenHeight);
    }
}
```

```
setLayout(null);

setBackground(new Color(18, 24, 32));

titleLabel = new Label("ENGINEERING STUDENT REGISTRATION
SYSTEM",Label.CENTER);

titleLabel.setFont(new Font("Arial", Font.BOLD, 30));

titleLabel.setForeground(new Color(0, 220, 255));

titleLabel.setBounds(0, 35, screenWidth, 50);

add(titleLabel);

subtitleLabel = new Label("Student Information and Academic Registration
Portal",Label.CENTER);

subtitleLabel.setFont(new Font("Arial", Font.PLAIN, 16));

subtitleLabel.setForeground(Color.LIGHT_GRAY);

subtitleLabel.setBounds(0, 85, screenWidth, 30);

add(subtitleLabel);

Panel mainPanel = new Panel();

mainPanel.setLayout(null);

mainPanel.setBackground(new Color(35, 45, 58));

mainPanel.setBounds(80, 130,screenWidth - 160,screenHeight - 230);

add(mainPanel);

int panelWidth = screenWidth - 160;

personalLabel = new Label("PERSONAL INFORMATION");

personalLabel.setFont(new Font("Arial", Font.BOLD, 20));

personalLabel.setForeground(new Color(0, 220, 255));

personalLabel.setBounds(40, 25, 350, 35);

mainPanel.add(personalLabel);

nameLabel = new Label("Full Name");

nameLabel.setForeground(Color.WHITE);

nameLabel.setFont(new Font("Arial", Font.PLAIN, 16));

nameLabel.setBounds(50, 80, 150, 30);

mainPanel.add(nameLabel);

nameField = new TextField();

nameField.setFont(new Font("Arial", Font.PLAIN, 16));

nameField.setBounds(200, 80, 300, 32);
```

```
mainPanel.add(nameField);

emailLabel = new Label("Email Address");
emailLabel.setForeground(Color.WHITE);
emailLabel.setFont(new Font("Arial", Font.PLAIN, 16));
emailLabel.setBounds(50, 130, 150, 30);
mainPanel.add(emailLabel);

emailField = new TextField();
emailField.setFont(new Font("Arial", Font.PLAIN, 16));
emailField.setBounds(200, 130, 300, 32);
mainPanel.add(emailField);

phoneLabel = new Label("Phone Number");
phoneLabel.setForeground(Color.WHITE);
phoneLabel.setFont(new Font("Arial", Font.PLAIN, 16));
phoneLabel.setBounds(50, 180, 150, 30);
mainPanel.add(phoneLabel);

phoneField = new TextField();
phoneField.setFont(new Font("Arial", Font.PLAIN, 16));
phoneField.setBounds(200, 180, 300, 32);
mainPanel.add(phoneField);

passwordLabel = new Label("Password");
passwordLabel.setForeground(Color.WHITE);
passwordLabel.setFont(new Font("Arial", Font.PLAIN, 16));
passwordLabel.setBounds(50, 230, 150, 30);
mainPanel.add(passwordLabel);

passwordField = new TextField();
passwordField.setEchoChar('*');
passwordField.setFont(new Font("Arial", Font.PLAIN, 16));
passwordField.setBounds(200, 230, 300, 32);
mainPanel.add(passwordField);

genderLabel = new Label("Gender");
genderLabel.setForeground(Color.WHITE);
genderLabel.setFont(new Font("Arial", Font.PLAIN, 16));
```

```
genderLabel.setBounds(50, 280, 150, 30);
mainPanel.add(genderLabel);
maleCheckbox = new Checkbox("Male");
maleCheckbox.setForeground(Color.WHITE);
maleCheckbox.setBackground(new Color(35, 45, 58));
maleCheckbox.setBounds(200, 280, 80, 30);
mainPanel.add(maleCheckbox);
femaleCheckbox = new Checkbox("Female");
femaleCheckbox.setForeground(Color.WHITE);
femaleCheckbox.setBackground(new Color(35, 45, 58));
femaleCheckbox.setBounds(300, 280, 100, 30);
mainPanel.add(femaleCheckbox);
academicLabel = new Label("ACADEMIC INFORMATION");
academicLabel.setFont(new Font("Arial", Font.BOLD, 20));
academicLabel.setForeground(new Color(0, 220, 255));
academicLabel.setBounds(600, 25, 350, 35);
mainPanel.add(academicLabel);
departmentLabel = new Label("Department");
departmentLabel.setForeground(Color.WHITE);
departmentLabel.setFont(new Font("Arial", Font.PLAIN, 16));
departmentLabel.setBounds(610, 80, 150, 30);
mainPanel.add(departmentLabel);
departmentChoice = new Choice();
departmentChoice.add("Select Department");
departmentChoice.add("Computer Science Engineering");
departmentChoice.add("Information Technology");
departmentChoice.add("Electronics and Communication");
departmentChoice.add("Electrical Engineering");
departmentChoice.add("Mechanical Engineering");
departmentChoice.add("Civil Engineering");
departmentChoice.setFont(new Font("Arial", Font.PLAIN, 14));
departmentChoice.setBounds(760, 80, 350, 35);
```

```
mainPanel.add(departmentChoice);

yearLabel = new Label("Academic Year");
yearLabel.setForeground(Color.WHITE);
yearLabel.setFont(new Font("Arial", Font.PLAIN, 16));
yearLabel.setBounds(610, 130, 150, 30);
mainPanel.add(yearLabel);

yearChoice = new Choice();
yearChoice.add("Select Year");
yearChoice.add("1st Year");
yearChoice.add("2nd Year");
yearChoice.add("3rd Year");
yearChoice.add("4th Year");
yearChoice.setFont(new Font("Arial", Font.PLAIN, 14));
yearChoice.setBounds(760, 130, 350, 35);
mainPanel.add(yearChoice);

skillsLabel = new Label("Technical Skills");
skillsLabel.setForeground(Color.WHITE);
skillsLabel.setFont(new Font("Arial", Font.PLAIN, 16));
skillsLabel.setBounds(610, 180, 150, 30);
mainPanel.add(skillsLabel);

javaCheckbox = new Checkbox("Java");
javaCheckbox.setForeground(Color.WHITE);
javaCheckbox.setBackground(new Color(35, 45, 58));
javaCheckbox.setBounds(760, 180, 70, 30);
mainPanel.add(javaCheckbox);

pythonCheckbox = new Checkbox("Python");
pythonCheckbox.setForeground(Color.WHITE);
pythonCheckbox.setBackground(new Color(35, 45, 58));
pythonCheckbox.setBounds(840, 180, 80, 30);
mainPanel.add(pythonCheckbox);

cppCheckbox = new Checkbox("C++");
cppCheckbox.setForeground(Color.WHITE);
```

```
cppCheckbox.setBackground(new Color(35, 45, 58));
cppCheckbox.setBounds(930, 180, 70, 30);
mainPanel.add(cppCheckbox);
termsCheckbox = new Checkbox("I agree to the terms and conditions");
termsCheckbox.setForeground(Color.WHITE);
termsCheckbox.setBackground(new Color(35, 45, 58));
termsCheckbox.setBounds(760, 230, 300, 30);
mainPanel.add(termsCheckbox);
registerButton = new Button("REGISTER");
registerButton.setFont(new Font("Arial", Font.BOLD, 16));
registerButton.setBackground(new Color(0, 160, 210));
registerButton.setForeground(Color.WHITE);
registerButton.setBounds(600, 290, 150, 45);
registerButton.addActionListener(this);
mainPanel.add(registerButton);
clearButton = new Button("CLEAR");
clearButton.setFont(new Font("Arial", Font.BOLD, 16));
clearButton.setBackground(new Color(90, 100, 110));
clearButton.setForeground(Color.WHITE);
clearButton.setBounds(770, 290, 150, 45);
clearButton.addActionListener(this);
mainPanel.add(clearButton);
exitButton = new Button("EXIT");
exitButton.setFont(new Font("Arial", Font.BOLD, 16));
exitButton.setBackground(new Color(180, 50, 50));
exitButton.setForeground(Color.WHITE);
exitButton.setBounds(940, 290, 150, 45);
exitButton.addActionListener(this);
mainPanel.add(exitButton);
resultTitleLabel = new Label("REGISTRATION STATUS");
resultTitleLabel.setFont(new Font("Arial", Font.BOLD, 18));
resultTitleLabel.setForeground(new Color(0, 220, 255));
```

```

resultTitleLabel.setBounds(40, 350, 300, 30);
mainPanel.add(resultTitleLabel);
resultLabel = new Label("Enter your details and click REGISTER.");
resultLabel.setFont(new Font("Arial", Font.PLAIN, 15));
resultLabel.setForeground(Color.WHITE);
resultLabel.setBounds(40, 390, panelWidth - 100, 80);
mainPanel.add(resultLabel);
addWindowListener(
    new WindowAdapter() {
        public void windowClosing(WindowEvent e) {
            dispose();
            System.exit(0);
        }
    });
setLocation(0, 0);
setVisible(true);
}

public void actionPerformed(ActionEvent e) {
    if (e.getSource() == registerButton) {
        String name = nameField.getText();
        String email = emailField.getText();
        String phone = phoneField.getText();
        String password = passwordField.getText();
        String department = departmentChoice.getSelectedItem();
        String year = yearChoice.getSelectedItem();
        String gender = "Not Selected";
        if (maleCheckbox.getState()) {
            gender = "Male";
        }
        if (femaleCheckbox.getState()) {
            gender = "Female";
        }
    }
}

```

```
String skills = "";
if (javaCheckbox.getState()) {
    skills += "Java ";
}
if (pythonCheckbox.getState()) {
    skills += "Python ";
}
if (cppCheckbox.getState()) {
    skills += "C++ ";
}
if (skills.equals("")) {
    skills = "None";
}
if (name.equals("") ||
    email.equals("") ||
    phone.equals("") ||
    password.equals("")) {
    resultLabel.setText("ERROR: Please fill in all personal information fields.");
    return;
}
if (department.equals("Select Department")) {
    resultLabel.setText("ERROR: Please select your department.");
    return;
}
if (year.equals("Select Year")) {
    resultLabel.setText("ERROR: Please select your academic year.");
    return;
}
if (!termsCheckbox.getState()) {
    resultLabel.setText(
        "ERROR: Please accept the terms and conditions.");
    return;
}
```



```

    }

    resultLabel.setText(
        "SUCCESS! Name: " + name +
        " | Email: " + email +
        " | Phone: " + phone +
        " | Gender: " + gender +
        " | Department: " + department +
        " | Year: " + year +
        " | Skills: " + skills
    );
}

if (e.getSource() == clearButton) {
    nameField.setText("");
    emailField.setText("");
    phoneField.setText("");
    passwordField.setText("");
    maleCheckbox.setState(false);
    femaleCheckbox.setState(false);
    javaCheckbox.setState(false);
    pythonCheckbox.setState(false);
    cppCheckbox.setState(false);
    termsCheckbox.setState(false);
    departmentChoice.select(0);
    yearChoice.select(0);
    resultLabel.setText("Enter your details and click REGISTER.");
}

if (e.getSource() == exitButton) {
    dispose();
    System.exit(0);
}
}

public static void main(String[] args) {

```

```

        new RegistrationForm();
    }
}

```

Output:

The screenshot shows the 'ENGINEERING STUDENT REGISTRATION SYSTEM' window. It features two main sections: 'PERSONAL INFORMATION' and 'ACADEMIC INFORMATION'. The 'PERSONAL INFORMATION' section includes fields for Full Name, Email Address, Phone Number, Password, and Gender (Male/Female). The 'ACADEMIC INFORMATION' section includes dropdowns for Department and Academic Year, checkboxes for Technical Skills (Java, Python, C++), and a checkbox for 'I agree to the terms and conditions'. At the bottom, there are three buttons: 'REGISTER' (blue), 'CLEAR' (grey), and 'EXIT' (red). Below the form, a 'REGISTRATION STATUS' section displays the message: 'Enter your details and click REGISTER.'

REGISTERED

The screenshot shows the same 'ENGINEERING STUDENT REGISTRATION SYSTEM' window, but now the form fields are populated with sample data: Full Name 'G Hasini', Email Address 'hasinigopu@gmail.com', Phone Number '9743645374', Password '*****', Gender 'Female', Department 'Computer Science Engineering', and Academic Year '3rd Year'. The 'REGISTER' button is highlighted in blue. Below the form, the 'REGISTRATION STATUS' section displays a success message: 'SUCCESS! Name: G Hasini | Email: hasinigopu@gmail.com | Phone: 9743645374 | Gender: Female | Department: Computer Science Engineering | Year: 3rd Year | Skills: Java'.

CLEAR

The screenshot shows the 'ENGINEERING STUDENT REGISTRATION SYSTEM' window with all form fields reset to their default state. The 'PERSONAL INFORMATION' section has empty fields for Full Name, Email Address, Phone Number, Password, and Gender. The 'ACADEMIC INFORMATION' section has dropdowns for Department and Academic Year, and checkboxes for Technical Skills. The 'REGISTER' button is highlighted in blue. Below the form, the 'REGISTRATION STATUS' section displays the message: 'Enter your details and click REGISTER.'

EXIT

Q2. Student Record System

Student Record System

Student ID:

Name:

Department:

Age:

Add

Update

Delete

Clear

Student ID	Name	Department	Age
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Student Record System

Student ID:

Name:

Department:

Age:

Add

Update

Delete

Clear

Student ID	Name	Department	Age
192425262	G Hasini	AIML	19
192425092	Harshi	AIDS	19

Q3. BASIC SHAPES

Code:

```
import java.awt.*;
import java.awt.event.*;

public class BasicShapes extends Frame implements ActionListener {
    Choice shapeChoice;
    Choice colorChoice;
    Button drawButton;
    Button clearButton;
    Button exitButton;
    DrawingCanvas canvas;
    Label titleLabel;
    Label statusLabel;

    public BasicShapes() {
        Dimension screen = Toolkit.getDefaultToolkit().getScreenSize();
        setTitle("Engineering Graphics - Basic Shapes");
        setSize(screen.width, screen.height);
        setLayout(null);
        setBackground(new Color(18, 24, 32));
        titleLabel = new Label("ENGINEERING GRAPHICS - BASIC SHAPES",
            Label.CENTER);
        titleLabel.setFont(new Font("Arial", Font.BOLD, 30));
        titleLabel.setForeground(new Color(0, 220, 255));
        titleLabel.setBounds(0, 35, screen.width, 50);
        add(titleLabel);
        Label subtitle = new Label("Interactive Shape Drawing and Color Selection",
            Label.CENTER);
        subtitle.setFont(new Font("Arial", Font.PLAIN, 16));
        subtitle.setForeground(Color.LIGHT_GRAY);

        subtitle.setBounds(0, 85, screen.width, 30);add(subtitle);

        Panel controlPanel = new Panel();
```

```
controlPanel.setLayout(null);
controlPanel.setBackground(new Color(35, 45, 58));
controlPanel.setBounds(100, 140, screen.width - 200, 130);
add(controlPanel);
Label shapeLabel = new Label("SELECT SHAPE");
shapeLabel.setFont(new Font("Arial", Font.BOLD, 15));
shapeLabel.setForeground(new Color(0, 220, 255));
shapeLabel.setBounds(80, 25, 130, 30);
controlPanel.add(shapeLabel);
shapeChoice = new Choice();
shapeChoice.add("Line");
shapeChoice.add("Rectangle");
shapeChoice.add("Oval");
shapeChoice.setFont(new Font("Arial", Font.PLAIN, 15));
shapeChoice.setBounds(220, 22, 180, 35);
controlPanel.add(shapeChoice);
Label colorLabel = new Label("SELECT COLOR");
colorLabel.setFont(new Font("Arial", Font.BOLD, 15));
colorLabel.setForeground(new Color(0, 220, 255));
colorLabel.setBounds(480, 25, 130, 30);
controlPanel.add(colorLabel);
colorChoice = new Choice();
colorChoice.add("Cyan");
colorChoice.add("Red");
colorChoice.add("Green");
colorChoice.add("Blue");
colorChoice.add("Yellow");
colorChoice.add("Black");
colorChoice.setFont(new Font("Arial", Font.PLAIN, 15));
colorChoice.setBounds(620, 22, 180, 35);
controlPanel.add(colorChoice);
drawButton = new Button("DRAW SHAPE");
```

```

drawButton.setFont(new Font("Arial", Font.BOLD, 14));
drawButton.setBackground(new Color(0, 160, 210));
drawButton.setForeground(Color.WHITE);
drawButton.setBounds(850, 20, 140, 40);
drawButton.addActionListener(this);
controlPanel.add(drawButton);

clearButton = new Button("CLEAR");
clearButton.setFont(new Font("Arial", Font.BOLD, 14));
clearButton.setBackground(new Color(90, 100, 110));
clearButton.setForeground(Color.WHITE);
clearButton.setBounds(1010, 20, 120, 40);
clearButton.addActionListener(this);
controlPanel.add(clearButton);

exitButton = new Button("EXIT");
exitButton.setFont(new Font("Arial", Font.BOLD, 14));
exitButton.setBackground(new Color(180, 50, 50));
exitButton.setForeground(Color.WHITE);
exitButton.setBounds(1150, 20, 120, 40);
exitButton.addActionListener(this);
controlPanel.add(exitButton);

canvas = new DrawingCanvas();
canvas.setBackground(Color.WHITE);
canvas.setBounds(100, 300, screen.width - 200, screen.height - 430);
add(canvas);

statusLabel = new Label("Select a shape and color, then click DRAW SHAPE.",
    Label.CENTER);
statusLabel.setFont(new Font("Arial", Font.BOLD, 15));
statusLabel.setForeground(new Color(0, 220, 255));
statusLabel.setBounds(100, screen.height - 110, screen.width - 200, 40);
add(statusLabel);
addWindowListener(

```

```

        new WindowAdapter() {
            public void windowClosing(WindowEvent e) {
                dispose();
                System.exit(0);
            }
        });
        setLocation(0, 0);
        setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        if (e.getSource() == drawButton) {
            String shape = shapeChoice.getSelectedItem();
            String color = colorChoice.getSelectedItem();
            canvas.setShape(shape);
            canvas.setShapeColor(color);
            canvas.repaint();

            statusLabel.setText("DRAWING: " + shape + " | COLOR: " + color);
        }

        if (e.getSource() == clearButton) {
            canvas.clearShape();
            canvas.repaint();
            statusLabel.setText("Drawing area cleared.");
        }

        if (e.getSource() == exitButton) {
            dispose();
            System.exit(0);
        }
    }

    class DrawingCanvas extends Canvas {
        String selectedShape = "";
        String selectedColor = "Cyan";

        public void setShape(String shape) {

```

```

        selectedShape = shape;
    }

    public void setShapeColor(String color) {
        selectedColor = color;
    }

    public void clearShape() {
        selectedShape = "";
    }

    public void paint(Graphics g) {
        g.setColor(Color.WHITE);
        g.fillRect(0, 0, getWidth(), getHeight());
        if (selectedShape.equals("")) {
            g.setColor(Color.GRAY);
            g.setFont(new Font( "Arial",Font.BOLD,22));
            g.drawString("DRAWING AREA",getWidth() / 2 - 80,getHeight() / 2);
            return;
        }
        g.setColor(getSelectedColor());
        if (selectedShape.equals("Line")) {
            g.drawLine(250, 120,getWidth() - 250,250);
        }
        else if (selectedShape.equals("Rectangle")) {
            g.drawRect(getWidth() / 2 - 200,100,400,220);
        }
        else if (selectedShape.equals("Oval")) {
            g.drawOval(getWidth() / 2 - 200,100,400,220);
        }
    }

    private Color getSelectedColor() {
        switch (selectedColor) {
            case "Red":
                return Color.RED;
        }
    }

```



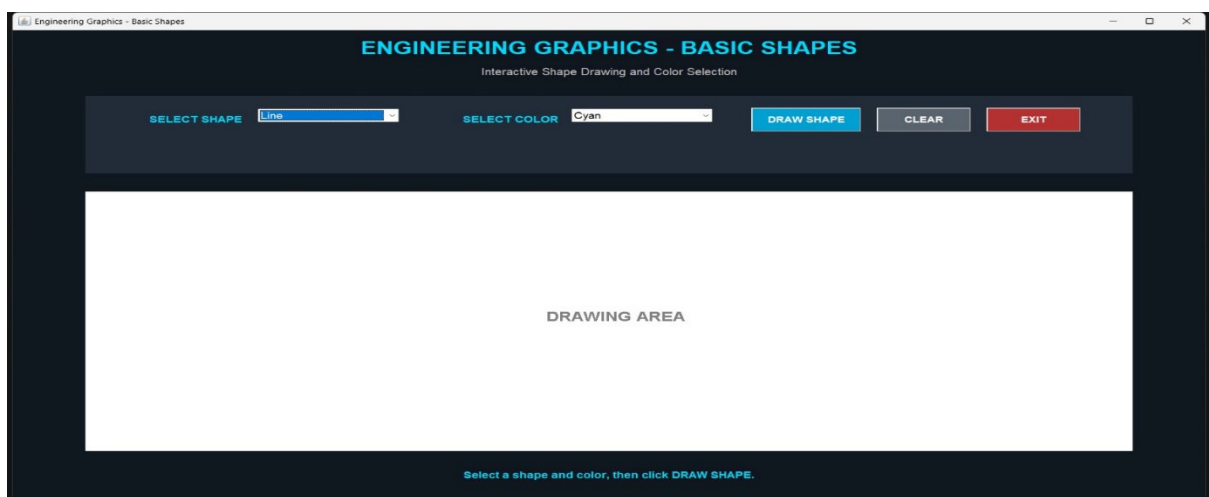
```

        case "Green":
            return Color.GREEN;
        case "Blue":
            return Color.BLUE;
        case "Yellow":
            return Color.YELLOW;
        case "Black":
            return Color.BLACK;
        case "Cyan":
        default:
            return Color.CYAN;
    }
}

public static void main(String[] args) {
    new BasicShapes();
}
}

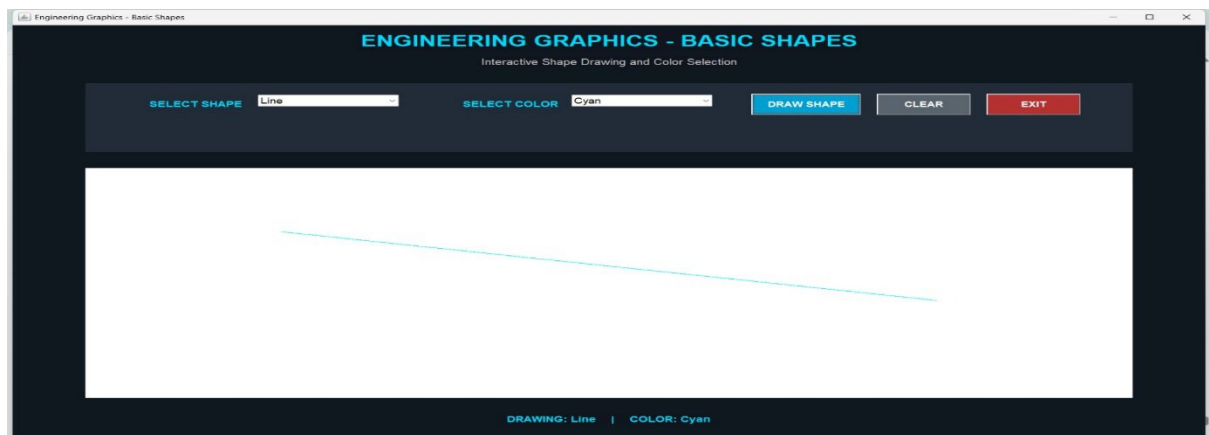
```

Output:



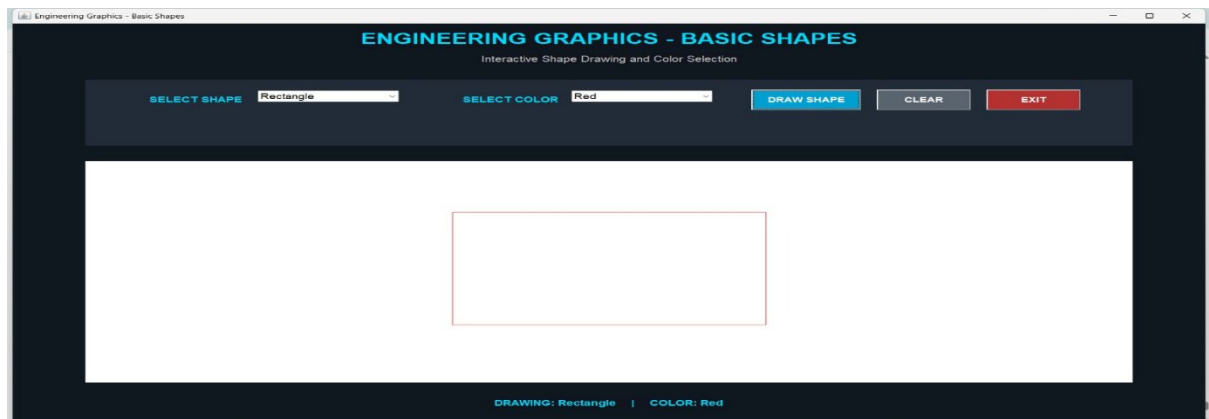
SELECT SHAPE: Line

SELECT COLOR: Cyan



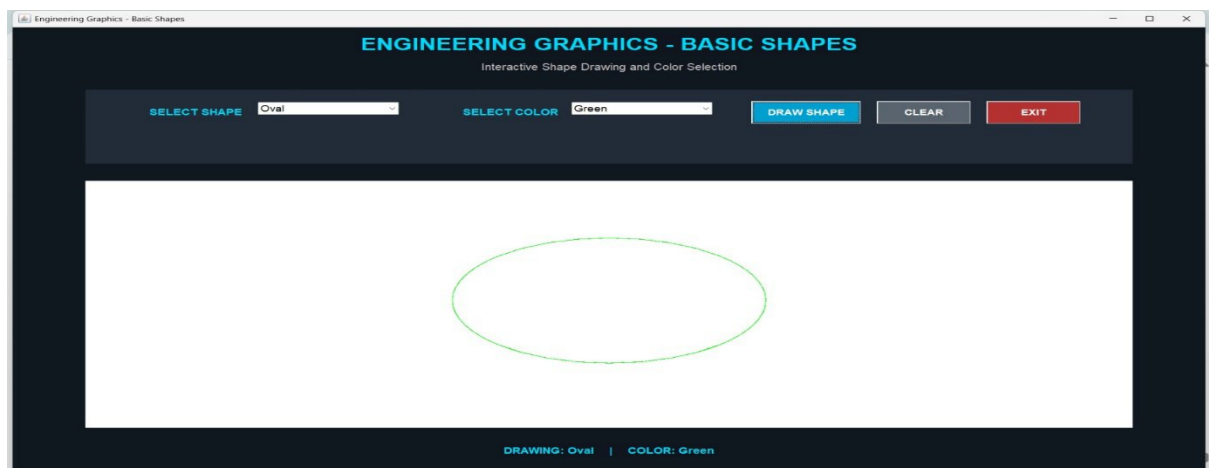
SELECT SHAPE: Rectangle

SELECT COLOR: Red



SELECT SHAPE: Oval

SELECT COLOR: Blue



CLEARED

Q4. Delegation Event Model

